

Mock test — Speed, energy and work: solutions

Self-marking. Give yourself the marks part by part, then add them up. Total 45 marks. The parts marked  are worth 27 marks, 60%: that is the pass mark. Read the common mistakes before you correct your own work: they show the wrong number you get when the method is wrong.

Exercise 1 — Multiple choice (8 marks, all)

Solution — 1-B, 2-C, 3-D, 4-A, 5-D, 6-A, 7-B, 8-C.

1. A stretched rubber band stores elastic (strain) energy.
2. $v = s/t = 150/2.0 = 75 \text{ km/h}$.
3. The distance does not change with time: the object is stationary.
4. Total energy is conserved. Potential energy decreases, kinetic energy and speed increase.
5. $E_k \propto v^2$: doubling v multiplies E_k by $2^2 = 4$.
6. A force moves the box through a distance. The other three cases have no movement.
7. $W = Fd$, so $1 \text{ J} = 1 \text{ N} \times 1 \text{ m}$.
8. $80 - 30 = 50 \text{ kJ}$. Wrong answer B: forgetting the thermal energy. Wrong answer D: taking the lost energy instead of what is left.

Exercise 2 — True or false (4 marks, all)

Solution — (a) F, (b) F, (c) T, (d) T.

(a) The gradient is the speed: steeper means faster. (b) Energy is transferred, never created or destroyed. (c) The height does not change, so $\Delta E_p = 0$: the push is horizontal. (d) Going up, speed drops and height grows.

Exercise 3 — Energy stores (3 marks, all)

Solution — Catapult: elastic (strain) $\rightarrow$ kinetic. Diver: chemical (in the muscles) $\rightarrow$ gravitational potential. Ball: gravitational potential $\rightarrow$ kinetic. Award the mark only if both stores are right.

Exercise 4 — A cycling journey (8 marks, 2)

Solution — (a)  [2] Stopped: B (horizontal line). Highest speed: C (steepest line). 1 mark each. (b) [2] Gradient of section C: vertical height = $60 - 20 = 40 \text{ km}$, horizontal width = $2.5 - 1.5 = 1.0 \text{ h}$.

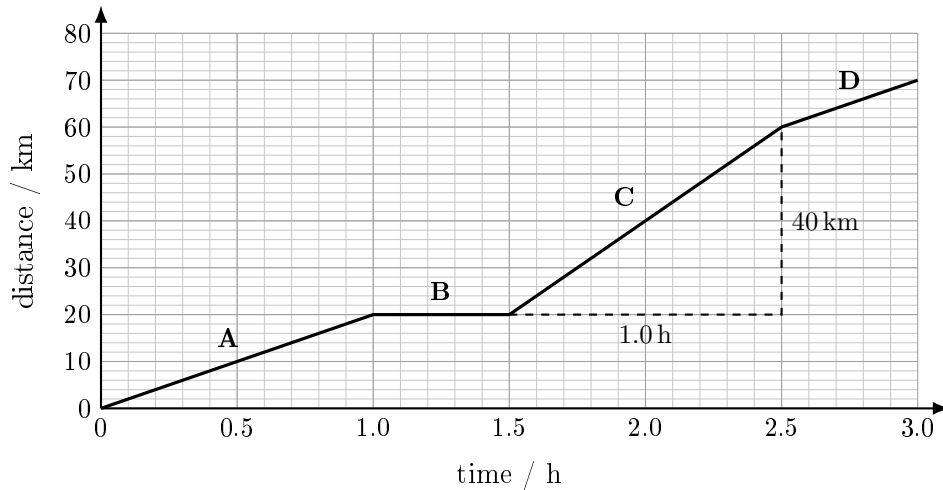
$$v = \frac{s}{t} = \frac{40 \text{ km}}{1.0 \text{ h}} = 40 \text{ km/h}$$

(c) [2] Total distance 70 km in total time 3.0 h, including the stop.

$$v_{\text{av}} = \frac{70 \text{ km}}{3.0 \text{ h}} \approx 23 \text{ km/h}$$

(d) [2]

$$v = \frac{40\,000 \text{ m}}{3600 \text{ s}} \approx 11 \text{ m/s}$$



Check and common mistakes — (c) Averaging the section speeds gives $(20 + 0 + 40 + 20)/4 = 20$ km/h: wrong, because the sections last different times. Leaving out the stop gives $70/2.5 = 28$ km/h: wrong, the stop counts. (d) Multiplying by 3.6 instead of dividing gives 144: a cyclist cannot go at 144 m/s. Check: 11 m/s is about 40 km/h.

Exercise 5 — The roller coaster (8 marks, 4 ⚡)

Solution — (a) ⚡ [2] Gravitational potential energy is transferred to kinetic energy as the trolley goes down. The principle of conservation of energy says that the energy lost from the first store equals the energy gained by the second (no friction). 1 mark for the stores, 1 for the principle.

(b) ⚡ [2]

$$mgh = \frac{1}{2}mv^2 \Rightarrow v^2 = 2gh \Rightarrow v = \sqrt{2gh}$$

The mass cancels. 1 mark for the equality, 1 for the final expression.

(c) [2]

$$\Delta E_p = mg\Delta h = (500)(10)(20) = 100\,000 \text{ J} = 100 \text{ kJ}$$

(d) [2] All of it becomes kinetic energy:

$$\begin{aligned} \frac{1}{2}mv^2 &= 100\,000 \Rightarrow v^2 = \frac{2 \times 100\,000}{500} = 400 \\ v &= \sqrt{400} = 20 \text{ m/s} \end{aligned}$$

Check and common mistakes — (c) Forgetting g gives 10 000 J. Using $h = 10$ m gives 50 kJ. (d) Forgetting the $\frac{1}{2}$ gives $v^2 = 200$, $v \approx 14$ m/s. Forgetting the square root gives 400 m/s. Check with (b): $\sqrt{2 \times 10 \times 20} = 20$ m/s.

Exercise 6 — Pushing a box up a ramp (8 marks, 4 ⚡)

Solution — (a) ⚡ [2] $W = Fd$: use 3.0 m, the distance moved along the direction of the force. $\Delta E_p = mgh$: use 0.90 m, the vertical height.

(b) [2]

$$W = Fd = (120)(3.0) = 360 \text{ J}$$

(c) [2] The weight is $mg = 300$ N.

$$\Delta E_p = (300)(0.90) = 270 \text{ J}$$

(d) ⚡ [2] The difference is $360 - 270 = 90$ J. It goes into the internal (thermal) energy store of the box and the ramp, because of friction between them. 1 mark for the store, 1 for friction.

Check and common mistakes — (b) Using 0.90 m gives 108 J. (c) Using 3.0 m gives 900 J, more than the work done by the girl: impossible without another source of energy. Check: the work done must be at least ΔE_p .

Exercise 7 — Stopping a car (6 marks, 2 🎯)

Solution — (a) 🎯 [1] $Fd = \frac{1}{2}mv^2$ (work done by the brakes = kinetic energy lost).
(b) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(1200)(15)^2 = 135\,000 \text{ J}$$

(c) [2]

$$d = \frac{E_k}{F} = \frac{135\,000}{6000} = 22.5 \text{ m}$$

(d) 🎯 [1] Four times as long: $E_k \propto v^2$ becomes four times bigger, and with the same force $d \propto E_k$.
Check: $d = 90 \text{ m}$.

Check and common mistakes — (b) Forgetting the square gives 9000 J. Forgetting the $\frac{1}{2}$ gives 270 000 J. (c) Dividing by the mass gives nonsense: units of distance need $\text{J/N} = \text{m}$.